4 DSA Patterns

= 250 LeetCode Problems Solved

A quick-reference sheet with the core idea behind each pattern, plus 10 practice questions per pattern to get you started.

1. Two Pointers

Use two indices moving through a data structure (often a sorted array) to avoid nested loops. Great for problems involving pairs, triplets, or comparing elements from opposite ends.

1. Two Sum II - Input Array Is Sorted
2. 3Sum
3. Container With Most Water
4. Valid Palindrome
5. Remove Duplicates from Sorted Array
6. Trapping Rain Water
7. Sort Colors (Dutch National Flag)
8. Move Zeroes
9. Squares of a Sorted Array
10. 4Sum

2. Sliding Window

Maintain a window (subarray/substring) that expands and shrinks based on a condition, avoiding recomputation. Ideal for 'longest/shortest/max subarray' type problems.

1. Best Time to Buy and Sell Stock
2. Longest Substring Without Repeating Characters
3. Longest Repeating Character Replacement
4. Permutation in String
5. Minimum Window Substring
6. Maximum Sum Subarray of Size K
7. Fruit Into Baskets
8. Sliding Window Maximum
9. Longest Subarray with Ones after Replacement
10. Find All Anagrams in a String

3. Binary Search

Not just for sorted arrays — use it whenever the search space can be halved based on a condition. Look for 'find minimum/maximum X such that condition holds' problems.

1. Binary Search (basic)
2. Search in Rotated Sorted Array
3. Find Minimum in Rotated Sorted Array
4. Find First and Last Position of Element in Sorted Ar
5. Koko Eating Bananas
6. Capacity to Ship Packages Within D Days
7. Median of Two Sorted Arrays
8. Search a 2D Matrix
9. Split Array Largest Sum
10. Peak Index in a Mountain Array

4. Dynamic Programming

Break a problem into overlapping subproblems and store results to avoid recomputation. Master 1D DP first (fibonacci-style), then move to 2D (grid/string comparison problems).

1. Climbing Stairs
2. House Robber
3. Longest Increasing Subsequence
4. Coin Change
5. Longest Common Subsequence
6. Edit Distance
7. 0/1 Knapsack (Partition Equal Subset Sum)
8. Word Break
9. Unique Paths
10. Longest Palindromic Substring